

179. Let the number be x .

$$\sqrt{(27 \div 5 \times x) + 15} = 5.4 \div 6 + 0.3$$

$$\Rightarrow \sqrt{\frac{(27 \div 5 \times x)}{15}} = \frac{5.4}{6} + 0.3$$

$$\Rightarrow \sqrt{\frac{27 \times x}{5 \times 15}} = \frac{5.4}{6} + 0.3$$

$$\Rightarrow \sqrt{\frac{9 \times x}{5 \times 5}} = 0.9 + 0.3$$

$$\Rightarrow \frac{3}{5} \sqrt{x} = 1.2$$

$$\Rightarrow \sqrt{x} = \frac{1.2 \times 5}{3} = 2 \Rightarrow x = (2)^2$$

$$\Rightarrow x = 2 \times 2 = 4^2$$

180. Given $\sqrt{24 \div 0.5 + 1} + \sqrt{18 \div 0.6 + 6}$

$$= \sqrt{24 \times \frac{1}{0.5} + 1} + \sqrt{18 \times \frac{1}{0.6} + 6}$$

$$= \sqrt{24 \times \frac{10}{5} + 1} + \sqrt{\frac{18 \times 10}{6} + 6}$$

$$= \sqrt{48 + 1} + \sqrt{30 + 6}$$

$$\sqrt{49} + \sqrt{36} = 7 + 6 = 13$$

181. Given $(\sqrt{63} + \sqrt{252}) \times (\sqrt{175} + \sqrt{28})$

$$= (\sqrt{63} + \sqrt{4 \times 63}) \times (\sqrt{25 \times 7} + \sqrt{4 \times 7})$$

$$= (\sqrt{7 \times 9} + \sqrt{4 \times 9 \times 7}) \times (\sqrt{5 \times 5 \times 7} + \sqrt{2 \times 2 \times 7})$$

$$= (3\sqrt{7} + 6\sqrt{7}) \times (5\sqrt{7} + 2\sqrt{7})$$

$$= 9\sqrt{7} \times 7\sqrt{7} = 441$$

182. Given $9x^2 + 25 - 30x$

$$\text{We have to find } \sqrt{9x^2 + 25 - 30x}$$

$$= \sqrt{(3x)^2 - 2 \cdot 3x \cdot 5 + (-5)^2} \quad \left\{ \because a^2 - 2ab + b^2 = (a - b)^2 \right\}$$

$$= \sqrt{(3x - 5)^2} = 3x - 5$$

183. $\sqrt{\frac{3}{11}} = \sqrt{\frac{3 \times 11}{11 \times 11}} = \frac{\sqrt{33}}{11}$

$$= \frac{5.745}{11} = 0.5223$$

184. Let the number be x .

$$\text{Given } \sqrt{x} + 14 = \sqrt{2601}$$

$$\text{or, } \sqrt{x} = 51 - 14 = 37$$

$$\text{or } x = 37^2 = 1369$$

185. $a = \frac{\sqrt{3}}{2}$ (given)

$$\therefore \sqrt{1+a} + \sqrt{1-a}$$

$$= \sqrt{1 + \frac{\sqrt{3}}{2}} + \sqrt{1 - \frac{\sqrt{3}}{2}}$$

$$= \sqrt{\frac{2+\sqrt{3}}{2}} + \sqrt{\frac{2-\sqrt{3}}{2}}$$

$$= \sqrt{\frac{2(2+\sqrt{3})}{4}} + \sqrt{\frac{2(2-\sqrt{3})}{4}}$$

$$= \sqrt{\frac{4+2\sqrt{3}}{4}} + \sqrt{\frac{4-2\sqrt{3}}{4}}$$

$$= \sqrt{\frac{3+1+2 \times \sqrt{3} \times 1}{2}} + \sqrt{\frac{3+1-2 \times \sqrt{3} \times 1}{2}}$$

$$\left\{ \begin{array}{l} \sqrt{3}^2 + (1)^2 - 2 \times \sqrt{3} \times 1 = (\sqrt{3} - 1)^2 \\ \therefore (\sqrt{3})^2 + (1)^2 + 2 \times \sqrt{3} \times 1 = (\sqrt{3} + 1)^2 \\ a^2 + b^2 + 2ab = (a + b)^2 \\ a^2 + b^2 - 2ab = (a - b)^2 \end{array} \right\}$$

$$= \frac{\sqrt{(\sqrt{3}+1)^2}}{2} + \frac{\sqrt{(\sqrt{3}-1)^2}}{2}$$

$$= \frac{\sqrt{3}+1+\sqrt{3}-1}{2}$$

$$= \frac{2\sqrt{3}}{2} = \sqrt{3}$$

186. Given $\frac{5+\sqrt{10}}{5\sqrt{5}-2\sqrt{20}-\sqrt{32}+\sqrt{50}}$

$$= \frac{5+\sqrt{10}}{5\sqrt{5}-2 \times 2\sqrt{5}-2 \times 2\sqrt{2}+5\sqrt{2}}$$

$$= \frac{5+\sqrt{10}}{5\sqrt{5}-4\sqrt{5}-4\sqrt{2}+5\sqrt{2}}$$

$$= \frac{5+\sqrt{10}}{\sqrt{5}+\sqrt{2}} = \frac{\sqrt{5}(\sqrt{5}+\sqrt{2})}{\sqrt{5}+\sqrt{2}} = \sqrt{5}$$

187. Given: $\frac{(0.75)^3}{1-0.75} + [0.75 + (0.75)^2 + 1]$

$$= \frac{(0.75)^2 \times 0.75}{0.25} + [0.75 + 0.5625 + 1]$$

$$= 0.5625 \times 3 + [0.75 + 0.5625 + 1]$$

$$= 1.6875 + 2.3125 = 4$$

$$\text{Square root of } 4 = 2$$

188. Given $\sqrt{10+2\sqrt{6}+2\sqrt{10}+2\sqrt{15}}$

$$= \sqrt{10+2 \times \sqrt{3} \times \sqrt{2}+2 \times \sqrt{2} \times \sqrt{5}+2 \times \sqrt{3} \times \sqrt{5}}$$

$$= \sqrt{2+3+5+2 \times \sqrt{2} \times \sqrt{3}+2 \times \sqrt{5} \times \sqrt{2}+2 \times \sqrt{5} \times \sqrt{3}}$$

$$= \sqrt{(\sqrt{2})^2 + (\sqrt{3})^2 + (\sqrt{5})^2 + 2 \times \sqrt{2} \times \sqrt{3} + 2 \times \sqrt{5} \times \sqrt{2} + 2 \times \sqrt{5} \times \sqrt{3}}$$

$$\left\{ \because a^2 + b^2 + c^2 + 2ab + 2bc + 2ca = (a + b + c)^2 \right\}$$

$$= \sqrt{(\sqrt{2} + \sqrt{3} + \sqrt{5})^2}$$

$$= (\sqrt{2} + \sqrt{3} + \sqrt{5})$$